

Laboratory Book



B.Sc Drive Technology

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Lab Rules and Safety

- No eating or drinking in the laboratory
- Keep walkways clear (of bags, coats, etc)
- Report any accident or breakage to the instructor immediately
- If in doubt about any verbal or written instruction, ask before carrying it out
- Do not wear personal audio devices and turn off cell phones
- Do not leave any objects loose on the equipment, such as screws, nuts, or washers.
- Do not physically interact with rotating object. If the motor needs to be interacted, first make sure the power supply is turned of and the motor is at a standstill.
- Get instructor approval before working on any equipment
- Have the appropriate documentation before troubleshooting electronic equipment.
- Don't rush your work as this increases the risk of problems.
- If any equipment is broken or not working to the level required by the manual/lecture please report it to the supervisor.
- Leave the laboratory as you have found it. Points will be taken from the group/individual who has left their test bench untidy.

If you are unsure about a procedure, ask for help **IF IN DOUBT, DON'T TOUCH/DO IT - ASK FOR HELP**

Part 1

Installation of a Drive System

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1.1. Introduction

This lab exercise focuses on the installation and the application of a variable speed drive power train using a Programmable Logical Controller (PLC), an inverter and an Induction Machine (IM).

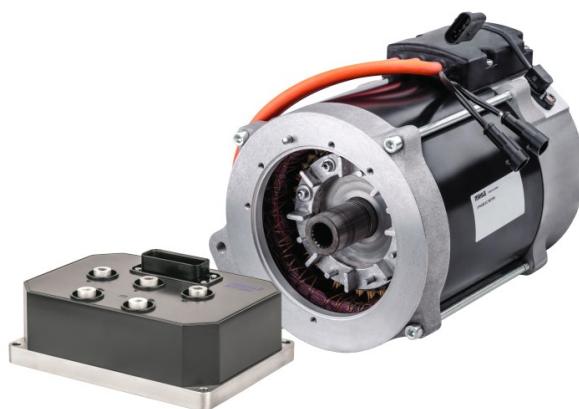


Figure 1.1.: Electric drives are becoming prominent in industry as converter efficiency and optimisation of manufacturing increases.

1.2. Learning Outcomes

During this exercise you will do all steps of the quick installation guide in given in [ABB document](#). The goal is to have a fully working drive power train in a no load configuration, which is controllable by the inputs of the PLC.¹

You will be familiar with the following topics after the exercise:

LO1 electrical installation of PLCs, inverters and IMs,

LO2 possible configurations for a drive power train with an IM,

LO3 configuration of an inverter with parameters.

For this lab you are to define sub-tasks for your groups to work efficiently during the time in the laboratory.

¹An industrial computer that is robust and adapted for the control of manufacturing processes, such as assembly lines, machines, robotic devices, or any activity that requires high reliability, ease of programming, and process fault diagnosis.

1.3. Literature Review

The following subsections cover information which may prove useful in understanding the underlying principles of this lab exercise.

Prior to attending the lab it is highly recommend to get accustomed to the ABB manual.

1.3.1. Inverters

Inverters, also called Alternating Current (AC) drives, or Variable Frequency Drive (VFD), are electronic devices that can turn Direct Current (DC) to AC. They are also responsible for controlling speed and torque for electric motors which are found in most devices we use to do work such as small electronics, transportation, and office appliances. As the lecture of this laboratory sessions have taught you, these motors need electricity to run. Matching the motor's speed to the required process is essential to avoid wasting energy. In factories, wasted energy and materials could put a business at risk, and so inverters are used to control electric motors, boosting productivity and saving energy.

An AC drive works between a power supply and an electric motor where power goes into the AC drive and regulates it. The regulated power is then sent to the motor.

An AC drive consists of a rectifier unit, a DC intermediate circuit, and an inverse conversion circuit. The rectifier unit inside an AC drive can be unidirectional or bidirectional. The former can accelerate and run the motor by taking energy from the electrical network. The latter can take the mechanical

rotation energy from the motor and send it back to the electrical system.

Before regulated power is received by the motor, it undergoes a conversion process inside the AC drive. The input power runs into a rectifier unit and the AC voltage is converted to DC voltage. The DC intermediate circuit dampens the DC voltage. It then flows through the inverse conversion circuit to convert the DC voltage back into AC voltage.

This process allows the AC drive to adjust the **frequency** and **voltage** supplied to the motor depending on the demands of process. The speed of the motor increases when the output voltage is at a higher frequency.

This means that the speed of the motor can be controlled via the operator interface.

1.4. Tasks

- During the preparation every group has defined sub-tasks and now it is time to work on these sub-tasks.
- As there are many different ways to structure the task into sub-tasks, it is up to the group to delegate the tasks.
- In general, there has to be an electrical installation of all the different modules required by the laboratory session, such as the PLC and the inverter together with the communication interface (MODBUS).²
- Prepare the cables for the connections (PLC-Inverter, Inverter-Machine, Inverter-Grid, etc.) along with the machine itself.
- The materials and the tools for the exercise will be provided by the lecturer or the laboratory supervisor (Manuel Berger, Ph.D), with every group having their own kit to work with.
- Make sure not to loose anything, since the modules will be used next year as well.

²A client/server data communications protocol in the application layer. It was originally designed for use with PLC, but has become a *de facto* standard communication protocol for communication between industrial electronic devices in a wide range of buses and networks.

NOTE: Cables will be manufactured by the group, if necessary.

Only the laboratory supervisor is allowed to turn-on the drive power train.

1.4.1. Transient Behaviour

- If the setup of the drive power train is completed, use the analogue output of the inverter to get a step response of the speed.

Find out how you can change the transient behaviour of the setup by changing parameters in the parameter list.

- Since there are many parameters, try to find out how they affect the transient response.
- Also think of possible scenarios, where certain parameter configurations could be reasonable.

1.5. Analysis and Report

No report has to be done for this exercise, but your performance during the exercise will be noted and will impact the final grade.

Part 2

Building a Lathe

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2.1. Introduction

The art of wood turning has become widespread for hobbyist carpenters over the last decades. The most common way to do wood turning is using a **lathe** which an example can be seen in **Fig. 2.1**.

A lathe is a machine tool which rotates a work piece¹ about an axis of rotation to perform various operations such as cutting, sanding, knurling,² drilling, deformation, facing, threading and turning, with tools that are applied to the work piece to create an object with symmetry around the axis of rotation.

The drive technology, which is often used in such lathes, is an **inverter** and an IM, especially for the ones which are on the more commercially affordable side. Sometimes gearboxes are also employed, but for this exercise the focus is on the drive power train without gearboxes.

The purpose of gearboxes is to transfer the motor power to the wheels as well as to reduce this power in order to achieve more torque and less speed.

¹i.e., a woodwork

²Knurling is a manufacturing process, typically conducted on a lathe, whereby a pattern of straight, angled or crossed lines is rolled into the material. Knurling can also refer to material that has a knurled pattern. While this process is generally used for metal working, knurling is also used in woodworking for hard wood.



Figure 2.1.: A Modern lathe used in industry [1]. It has been brought up to the author that the image presented here does not technically fit to a lathe used for wood-working. However, for aesthetic reasons, the image is kept as it is still a lathe. The determination of the technical mistakes this image brings to the script is left to the reader as a gentle exercise.

2.2. Learning Outcomes

During this lab exercise you will develop a strategy to design and realise the drive power train of a lathe. The goal is to have a fully working drive power train for a lathe, which is manually controllable.

The learning outcomes of this lab exercises are:

- LO1** Developing strategies for easy application for inverters and IMs,
- LO2** Using the input and output of an inverter to control via external methods,
- LO3** How to configure an inverter.

2.3. Lab Preparation

For this exercise the preparation does **NOT** require extensive literature review like other laboratory sessions do and requires of you only the following research:

1. Investigate reasonable wood turning speeds with a diameter d between 130 mm to 500 mm.
2. How is the turning speed dependent to the diameter of the wood as a function of diameter.

3. How would you distribute six (6) discrete turning speeds in the range, which came from the investigations?

2.4. Tasks

During this lab exercise the first task will be to set up the inverter and the induction drive similar to what was done in [Laboratory Exercise 1](#), but [without](#) the PLC.

This configuration will be the starting point for this exercise.

Once this setup is attained there will be three (3) approaches to do.

2.4.1. First Task

The first task is to build a drive power train which has all the fundamental features of a typical lathe. These features are:

1. Seven (7) discrete turning speeds,
2. Signalisation of errors and warnings of the inverter,
3. An ☐ ON/OFF switch.

2.4.2. Second Task

The second task should build on top of the first task with additional features with the additional features being:

1. Three (3) discrete variable turning speed,
2. Signalisation of errors and warning of the inverter,
3. Additional analogue input to vary the turning speed continuously,
4. An ☐ ON/OFF switch,
5. As an output, the turning speed has to be provided as a current between 4 mA to 20 mA,
6. With an input it should be possible to change between type sets of dynamic parameters for the drive system.

2.4.3. Third Task

The third and the final approach is now up to the group and **optional**. At this point the drive power train can be changed based on the requirements of the group.

2.5. Analysis and Report

As this exercise has a part which is about realising the given goals, this has to be part of the report. If your group was not able to achieve the goals, write about which part of the goals were achieved and which were not.

Also include the reasons why your group were not able to realise the experiment. Make sure to state what you have done, and why you have done it.

What to Hand in

Your lab report, with your findings **must** be submitted to SAKAI no later than **2 weeks** after the experiment is concluded. Your lab report should roughly include the following points:

Topic	Description
Cover Page	The cover page needs to include: Title of the Experiment Student Name(s) Student Number(s) Module
Abstract	A short description of your work where you summarise your results in 100 - 200 words.
Introduction	The point in which you can hook the reader into reading your report. Start in with a broad picture and as you continue funnel the reader into what is relevant to your work. Here also include the literature research done for the lab preparation.
Materials and Methods	Explain to the reader what tools were used and give enough information for the work to be reproducible.
Results	Provide the reader with all the information gathered from the lab results and present them in a table of figure format.
Discussion and Conclusion	Considers whether the data you obtained support the hypothesis and discuss the ramifications of your results.
References	Present all the relevant references for your work. As the work is carried out in the electrical engineering, use the IEEE citation format.

Table 2.1.: A brief look on the stages of writing a lab report.

If you need a refresher or would like to learn more about scientific lab report writing, please look at the Appendix.

Part 3

Synchronous and Induction Drives

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3.1. Introduction

This lab exercise will cover two (2) types of electrical drives used in industry:

1. IM in motor operation,¹
2. Synchronous Machine (SynM) in generator operation.

¹This is also known as asynchronous machine.

These machines' properties will be analysed on a typical test bench. In an industrial setting, machines have to be tested for specific applications, and depending on the application² specific tests would be conducted. Possible tests could be for life cycle of the machine, analysing its equivalent circuit, determining inrush current, etc..

²i.e., lifts, linear motion, generator.

The test bench for this exercise is a so called **power test bench**, with sensors to determine the parameters (torque, power factor, temperature, efficiency, rotation speed) for a machine operating in normal operation.

3.2. Learning Outcomes

During the lab exercises you will work on IM and SynM. The learning outcomes are as follows:

- LO1** Understanding the IM behaviour,
- LO2** Working with IMs and testing its parameters,
- LO3** Understanding SynM behaviour,
- LO4** Working with SynMs and testing its parameters,

The aim is to understand their working principle and to develop an understanding of their behaviour using graphs, which usually are provided in data sheets.

- First task will focus on IM and determining its characteristics.
 - Once determined, a compensation network will be integrated to the system to re-evaluate its power factor ($\cos \phi$).
- Second tasks will be study of a SynM.
 - Operating in generator mode, you will feed DC current to the excitation windings to generate voltage across the windings and plot it (x: DC Excitation, y: Single Phase),
 - Once the curve is obtained, you will introduce a 3-phase load to the generator windings and under a constant DC excitation, plot winding voltage (x: Resistor value, y: Single Phase).

3.3. Literature Research

Before setting foot into the lab, it is important to know some fundamental knowledge on the drives you will be experimenting in.

Here are some topics you should have knowledge of:

- Know the general working principle of an IM,
- Understand how compensation networks work, and how can an IM work with a compensator to achieve a power factor of 1 (i.e., $\cos \phi = 1$).
- Know the general working principle of a SynM.

It is **recommended** to read the following sections **prior** to attending the lab session.

3.4. Literature Review

3.4.1. Induction Drive

In an IM, AC power supplied to drive's stator creates a magnetic field (B) which rotates **in synch** with the AC grid. An IM's rotor rotates at a slower speed than the stator field. The IM stator's magnetic field is therefore *changing or rotating relative to the rotor*. This induces an opposing current in the rotor, in effect the motor's secondary winding. The rotating magnetic flux induces currents in the rotor windings, in a manner similar to currents induced in a transformer's secondary winding(s).

The induced currents in the rotor windings in turn create magnetic fields in the rotor that *react against the stator field*. The direction of the rotor magnetic field opposes the change in current through the rotor windings, following Lenz's law. The cause of induced current in the rotor windings is the rotating stator magnetic field, so to oppose the change in rotor-winding, currents the rotor turns in the direction of the stator magnetic field.

The rotor accelerates until the magnitude of induced rotor current and torque balances the load on the rotor. Since rotation at synchronous speed does not induce rotor current, an induction motor always operates slightly slower than synchronous speed. The difference, or **slip** between actual and synchronous speed varies from about 0.5% to 5.0% for an average motor. The IM's essential character is that **torque is created solely by induction** instead of the rotor being separately excited as in synchronous or DC machines or being self-magnetised as in Permanent Magnet (PM) motors.

For rotor currents to be induced, the speed of the physical rotor must be lower than that of the stator's rotating magnetic field (n_s); otherwise the magnetic field would not be moving relative to the rotor conductors and no currents would be induced. As the speed of the rotor drops below synchronous speed, the rotation rate of the magnetic field in the rotor increases, inducing more current in the windings and creating more torque.

The ratio between the rotation rate of the magnetic field induced in the rotor and the rotation rate of the stator's rotating field is called **slip**.

Under load, the speed drops and the slip increases enough to create sufficient torque to turn the load. For this reason, IMs are sometimes referred to as **asynchronous motors**.

3.4.2. Synchronous Drive

A SynM is an AC electric motor in which, at steady state, the rotation of the shaft is **synchronised with the frequency of the supply current** [2].

The rotation period is exactly equal to an integral number of AC cycles.

SynMs use electromagnets as the stator of the motor which create a magnetic field that rotates in time with the oscillations of the current. The rotor with permanent magnets or electromagnets turns in step with the stator field at the same rate and as a result, provides the second synchronised rotating magnet field.

A SynM is termed doubly fed if it is supplied with independently excited multi-phase AC electromagnets on both the rotor and stator.

3.4.3. Synchronous Speed

The synchronous speed of a SynM is given:

$$n_s = 60 \frac{f}{P} = 120 \frac{f}{p}, \quad \text{or} \quad \omega_s = 2\pi \frac{f}{P} = 4\pi \frac{f}{p}.$$

where f is the frequency of the AC (Hz), p is the magnetic poles, and P is the number of pole pairs with the relation:

$$P = \frac{p}{2}$$

Starting Methods

Above a certain size, SynMs cannot self-start.

This property is due to rotor inertia as it cannot instantly follow the stator's magnetic field.

Since a SynM produces no inherent average torque at standstill, it cannot accelerate to synchronous speed **without a supplemental mechanism**.

Large motors operating on commercial power include a squirrel-cage induction winding that provides sufficient torque for acceleration and also serves to damp motor speed oscillations.

Once the rotor nears the synchronous speed, the field winding becomes excited and the motor pulls into synchronisation. Very large motor systems may include a “pony” motor that accelerates the unloaded SynM before load is applied. Electronically controlled motors can be accelerated from zero speed by changing the frequency of the stator current.

Small SynMs are commonly used in line-powered electric mechanical clocks or timers that use the power line frequency to run the gear mechanism at the correct speed. Such small SynMs are able to start without assistance if the moment of inertia of the rotor and its mechanical load are sufficiently small.

The motor accelerates from slip speed to synchronous speed during an accelerating half cycle of the reluctance torque. Single-phase SynMs such as in electric wall clocks can freely rotate in either direction, unlike a shaded-pole type.



Figure 3.1.: Synchronous condenser installation at Templestowe substation, Melbourne, Victoria, Australia. Built by ASEA in 1966, the unit is hydrogen cooled and capable of three phase power at 125 MVA [3].

3.4.4. Static VAR Compensator (SVC)

A Static VAR Compensator (SVC) is a set of electrical devices for providing fast-acting reactive power on high-voltage electricity transmission networks [4]. SVCs are part of the flexible AC transmission system device family, regulating voltage, power factor, harmonics and stabilising the system. A static VAR compensator has no significant moving parts.³

³other than internal switch-gear.

Prior to the invention of the SVC, power factor compensation was the sole domain of large rotating machines such as synchronous condensers or switched capacitor banks with an example shown in **Fig. 3.1.**

The SVC is an automated impedance matching device, designed to bring the system closer to unity power factor. SVCs are used in two (2) main situations:

Transmission SVC Connected to the power system, to regulate the transmission voltage.

Industrial SVC Connected near large industrial loads, to improve power quality.

In transmission applications, the SVC is used to regulate the grid voltage. If the power system's reactive load is capacitive (leading), the SVC will use thyristor controlled reactors to consume VARs from the system, lowering the system voltage. Under inductive (lagging) conditions, the capacitor banks are automatically switched in, thus providing a higher system voltage. By connecting the thyristor-controlled reactor, which is continuously variable, along with a capacitor bank step, the net result is continuously variable leading or lagging power. In industrial applications, SVCs are typically placed near high and rapidly varying loads, such as arc furnaces, where they can smooth flicker voltage.

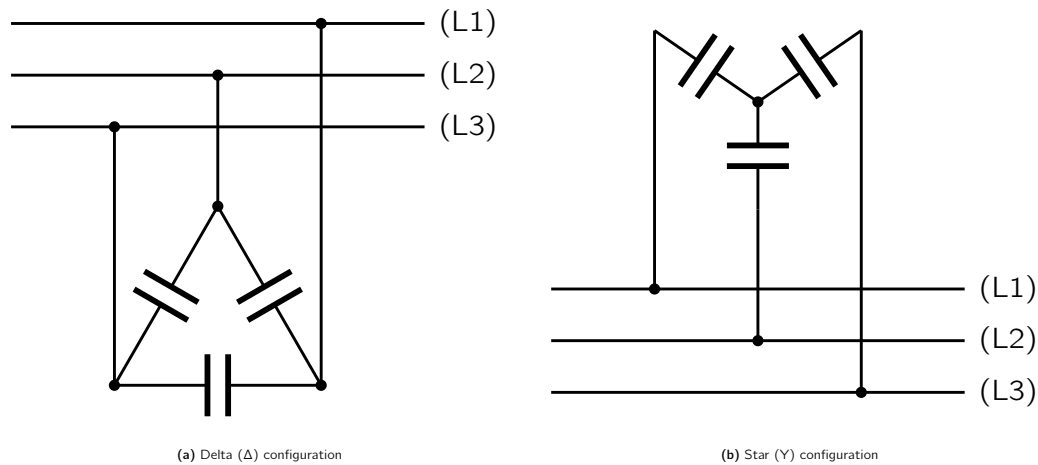


Figure 3.2.: Types of compensation network connections.

You can see the possible 3-phase configuration of a compensation network in **Fig. 3.2**.

3.5. Tasks

Below are the tasks for the two (2) lab sessions pertaining to electric drives.

3.5.1. Induction Drive

For the IM all the tasks should be done in both **Star** and **Delta** configuration shown in **Fig. 3.3**.

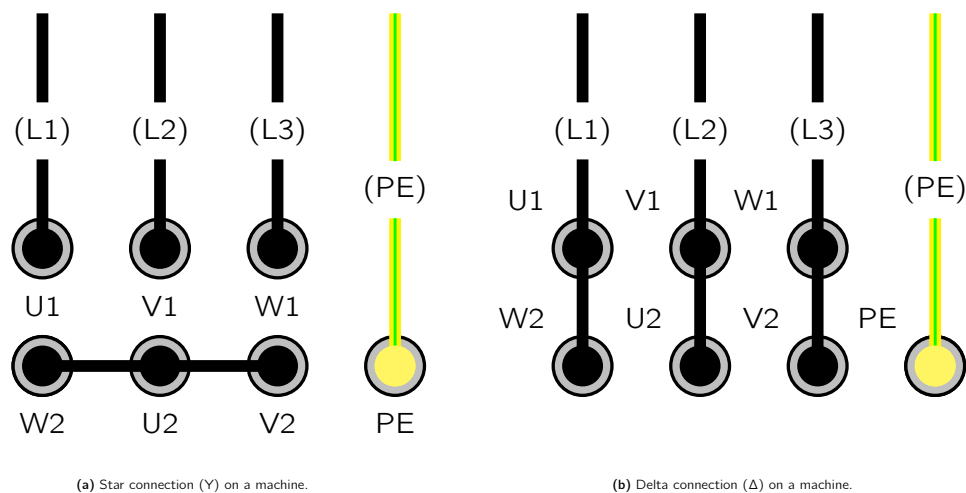


Figure 3.3.: Types of connections for electric motors.

Only the supervisor or lecturer is allowed to operate the test bench.
Failure to follow this direction will result in immediate fail of the lab.

Each task develops over the previous one, therefore try to do the tasks in this order:

1. Set up the test bench connections shown in **Fig. 3.4**,
2. Using the test bench, measure the no load speed (n_0),
3. Measure the speed-torque behaviour during normal operation using the computer,
4. Measure the torque (T), the power factor ($\cos \phi$) and efficiency (η) with respect to rpm (n).
5. Calculate and apply the compensation network of the induction machine.
6. Redo step 4 with the compensation applied to the motor. Use the diagram in **Fig. 3.5**.

3.5.2. Synchronous Generator

For the synchronous generator **only the star configuration** should be applied.

1. Remove the IM and its braking unit and replace it with the synchronous generator and its breaking unit,
2. Supply the excitation winding (**F1** and **F2**) with the variable DC source of the supply unit.

Make sure, the voltage is zero before you start. Use **Fig. 3.6** as the reference for your connections.

3. Run the generator via the brake with the nominal speed, and measure the induced voltage using a multimeter using the terminals (e.g., **U1** and **U2**).
4. Apply a 3-phase load to the generator windings and under a constant DC excitation measure the winding voltage. Use **Fig. 3.7** as the reference for your connections.

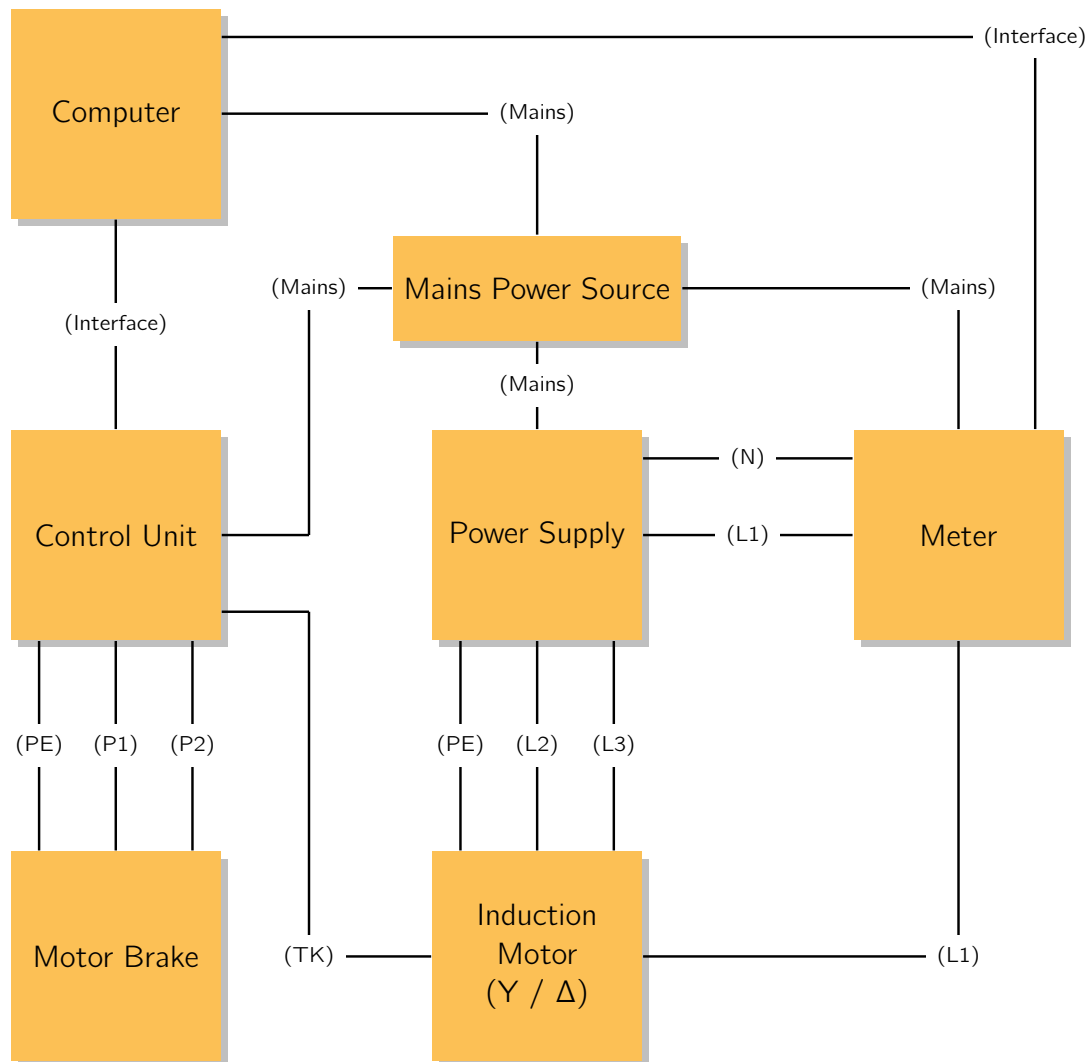


Figure 3.4.: The connection Diagram for Induction drive exercise 1.

When you are plugging the control unit interface to the computer, make sure you plug it to the correct driver slot (i.e., from the backside view, it is the port on the right hand side) and make sure the dial is set to **INTERFACE**, int. /ext. switch to **int.** and int./INTERFACE on **INTERFACE**.

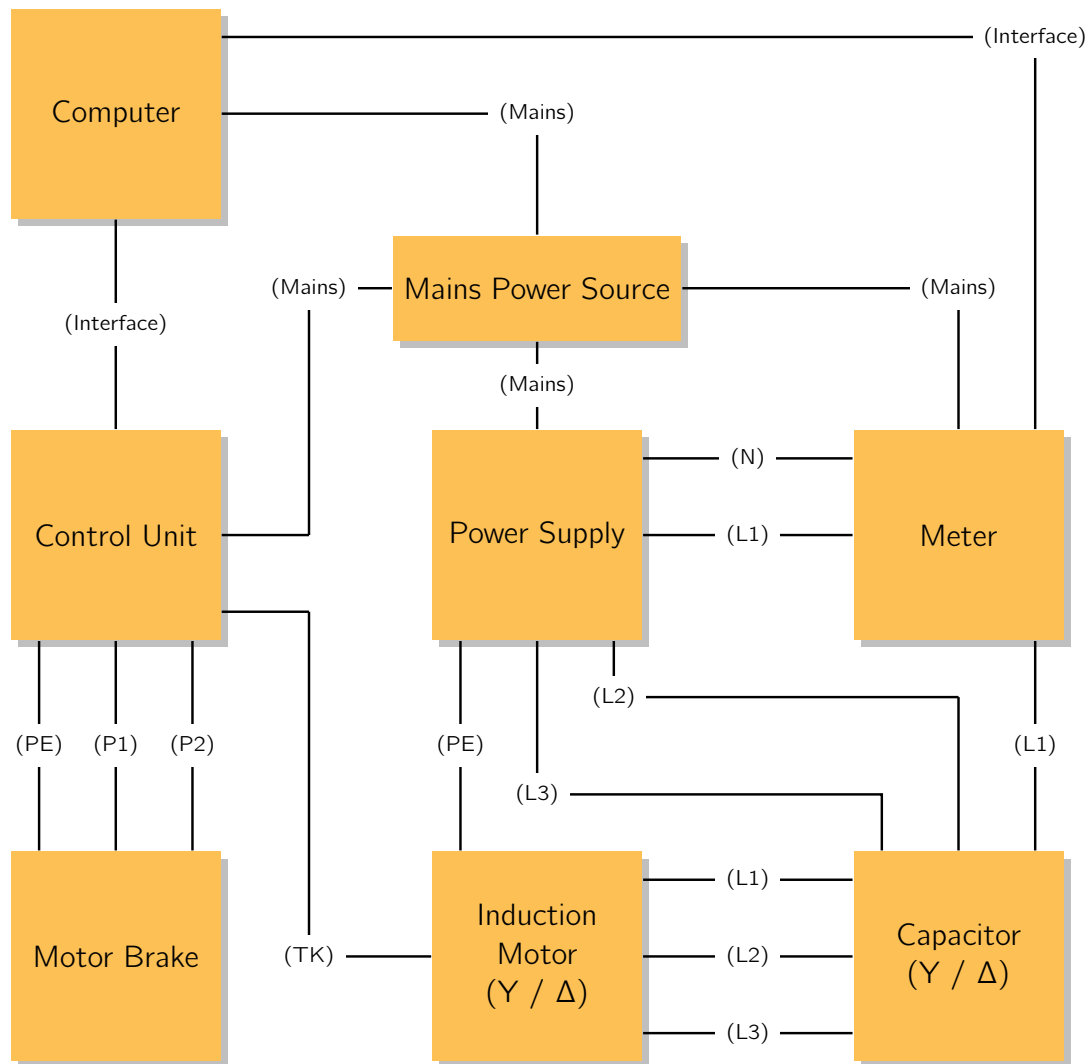


Figure 3.5.: The connection Diagram for Induction drive exercise 2.

When you are plugging the control unit interface to the computer, make sure you plug it to the correct driver slot (i.e., from the backside view, it is the port on the right hand side) and make sure the dial is set to **INTERFACE**, int. /ext. switch to **int.** and int./INTERFACE on **INTERFACE**.

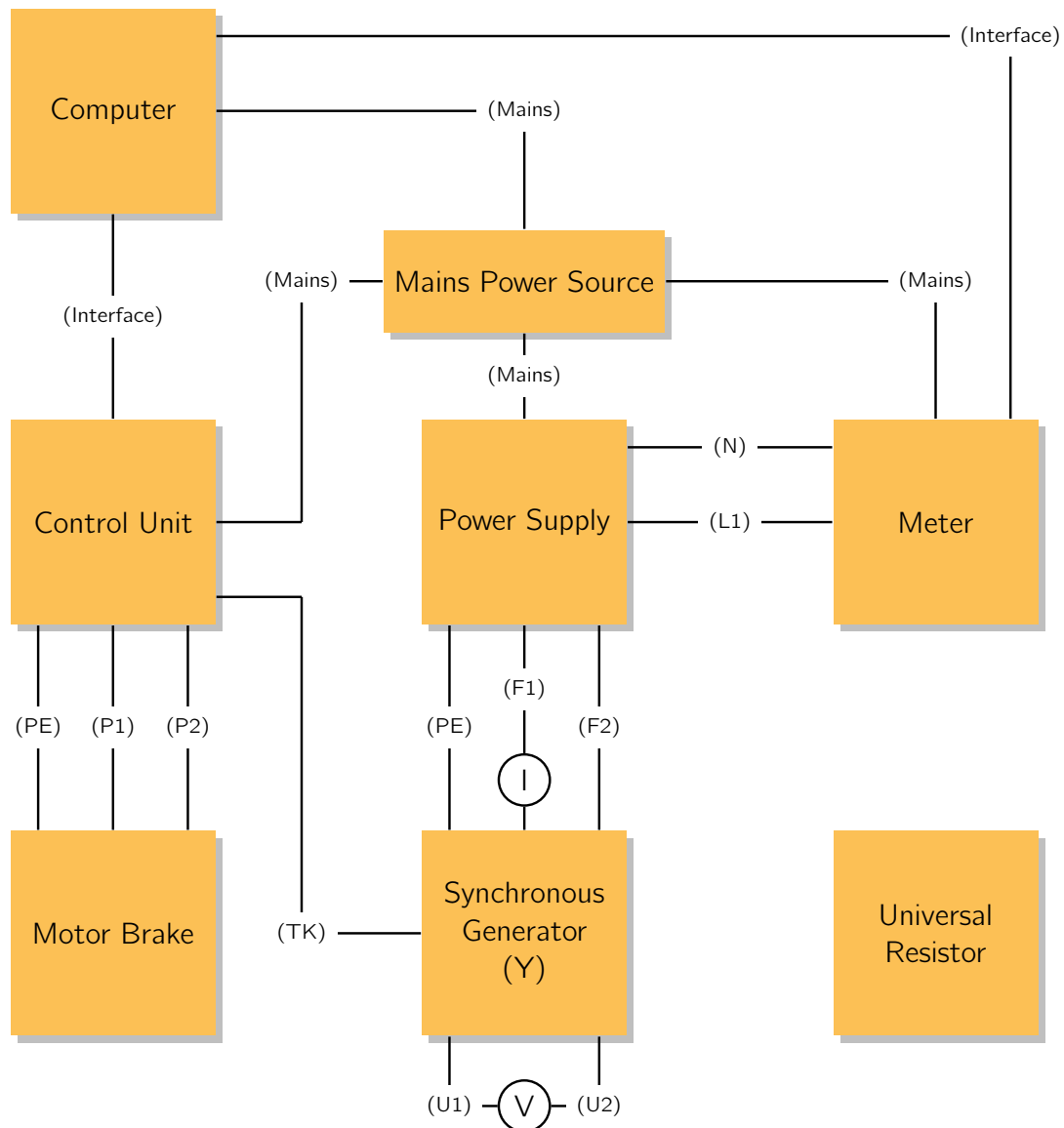


Figure 3.6.: The connection Diagram for Synchronous drive exercise 1.

When you are plugging the control unit interface to the computer, make sure you plug it to the correct driver slot (i.e., from the backside view, it is the port on the right hand side), Make sure the dial is set to **MANUAL** To start/stop the motor press the START/STOP button.

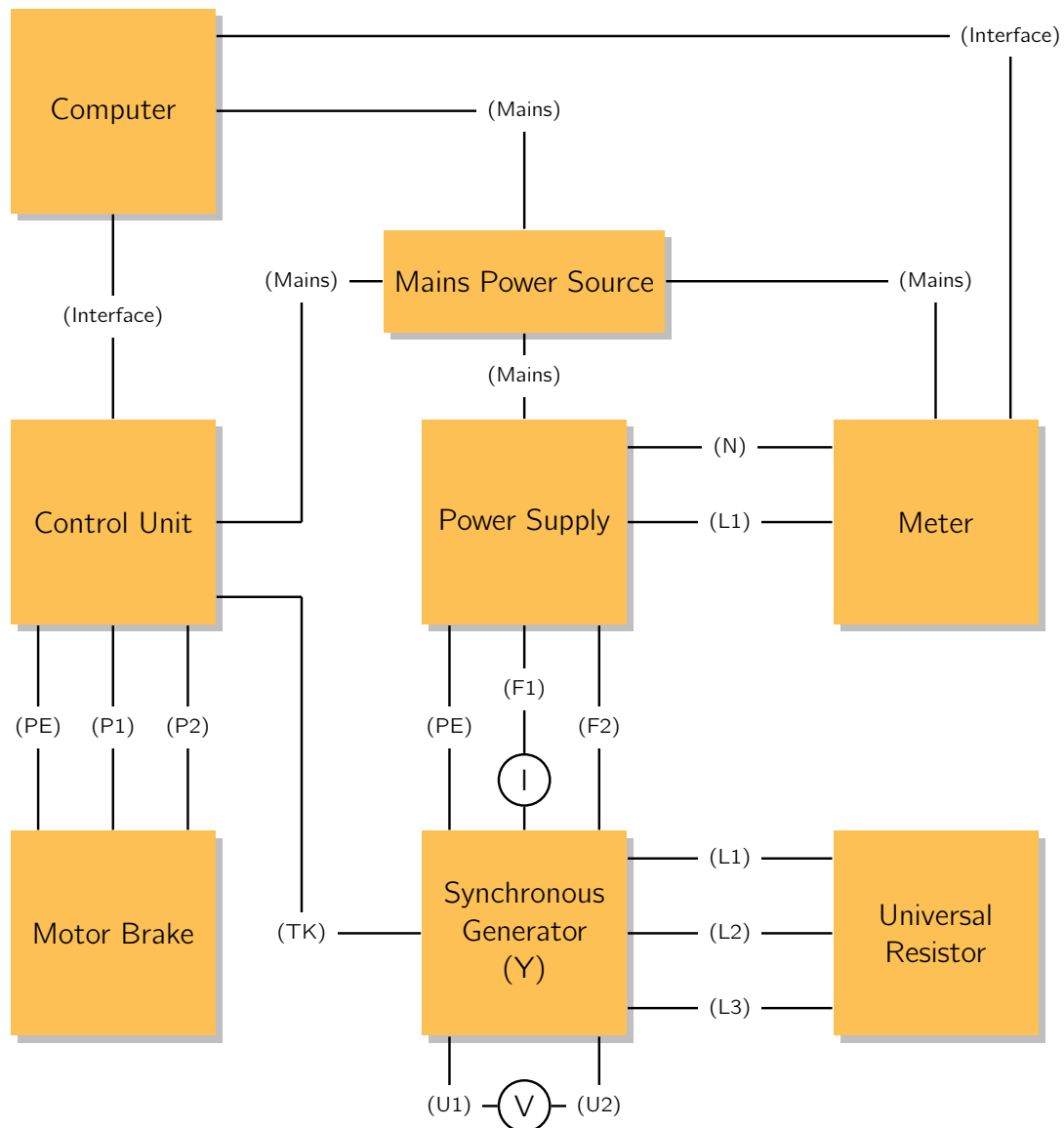


Figure 3.7.: The connection Diagram for Synchronous drive exercise 2.

- When you are plugging the control unit interface to the computer, make sure you plug it to the correct driver slot (i.e., from the backside view, it is the port on the right hand side),
- Make sure the dial is set to **MANUAL**
- To start/stop the motor press the START/STOP button.

What to Hand in

Your lab report, with your findings **must** be submitted to SAKAI no later than **2 weeks** after the experiment is concluded. Your lab report should roughly include the following points:

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Part 4

Photovoltaic Cells

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4.1. Introduction

As the times are changing, so does the source of our energy. The industrial revolution has enabled the use of coal and other types of fossil fuels to power the machinery's which built the modern world. Currently we are in the midst of a transition of energy sources to something more renewable, and nothing is more renewable than the energy produced by the great fireball in the sky.

To harness this energy, solar panels are being deployed which are used from charging a pocket calculator to feeding entire cities. This lab exercise will see you working with a small setup to show the working principle of photovoltaics in an island configuration and in connection with the grid [5].

4.2. Learning Objectives

This lab focuses on the use of solar cells in island and grid configuration together with a buffer (i.e., battery), network activators, inverters, chargers and transformers.

After the successful completion of the exercise, you will be familiar with:

- LO1** Power and efficiency of the solar cell modules,
- LO2** Supplying loads in an island configuration,
- LO3** Supplying loads and feeding back energy to the grid.

4.3. Lab Preparation

As the topic of solar cells are not covered in the **B.Sc Drive Technology** course, given its focus is on the aspect of machine design, some research will be required by the students to get the most out of the lab. The major topic suggest to students are:

- Understanding mono-, polycrystalline and thin-film panels,
- A brief literature survey on how to measure the maximum power,
- Have a read on what an island and grid configuration is.

4.4. Literature Review

4.4.1. The Electrical Grid

An electrical grid is an interconnected network for delivering electricity from producers¹ to consumers producers² [6]. In general, electrical grids consist of numerous power stations, electrical substations to step voltage up or down, electric power transmission to carry power over long distances, and finally electric power distribution to customers [8].

¹i.e., coal power-plants, hydro-dams

²i.e., manufacturers, residential households

In its last step of distribution, voltage is stepped down one last time to the required service voltage. Power stations are typically built close to energy sources and far from densely populated areas. This is generally done to mitigate environmental hazard affecting residential areas and faster energy generation from source to generator.

Electrical grids vary in size and can cover whole countries or continents (i.e., the European grid). From small to large there are micro-grids, wide area synchronous grids, and super grids. The combined transmission and distribution network is part of electricity delivery, known as the **power grid**.

Grids are almost nearly always **synchronous**. This means all distribution areas operate with three phase AC frequencies synchronised.³ This allows transmission of AC power throughout the area, connecting the electricity generators with consumers.

³so that voltage swings occur at almost the same time.

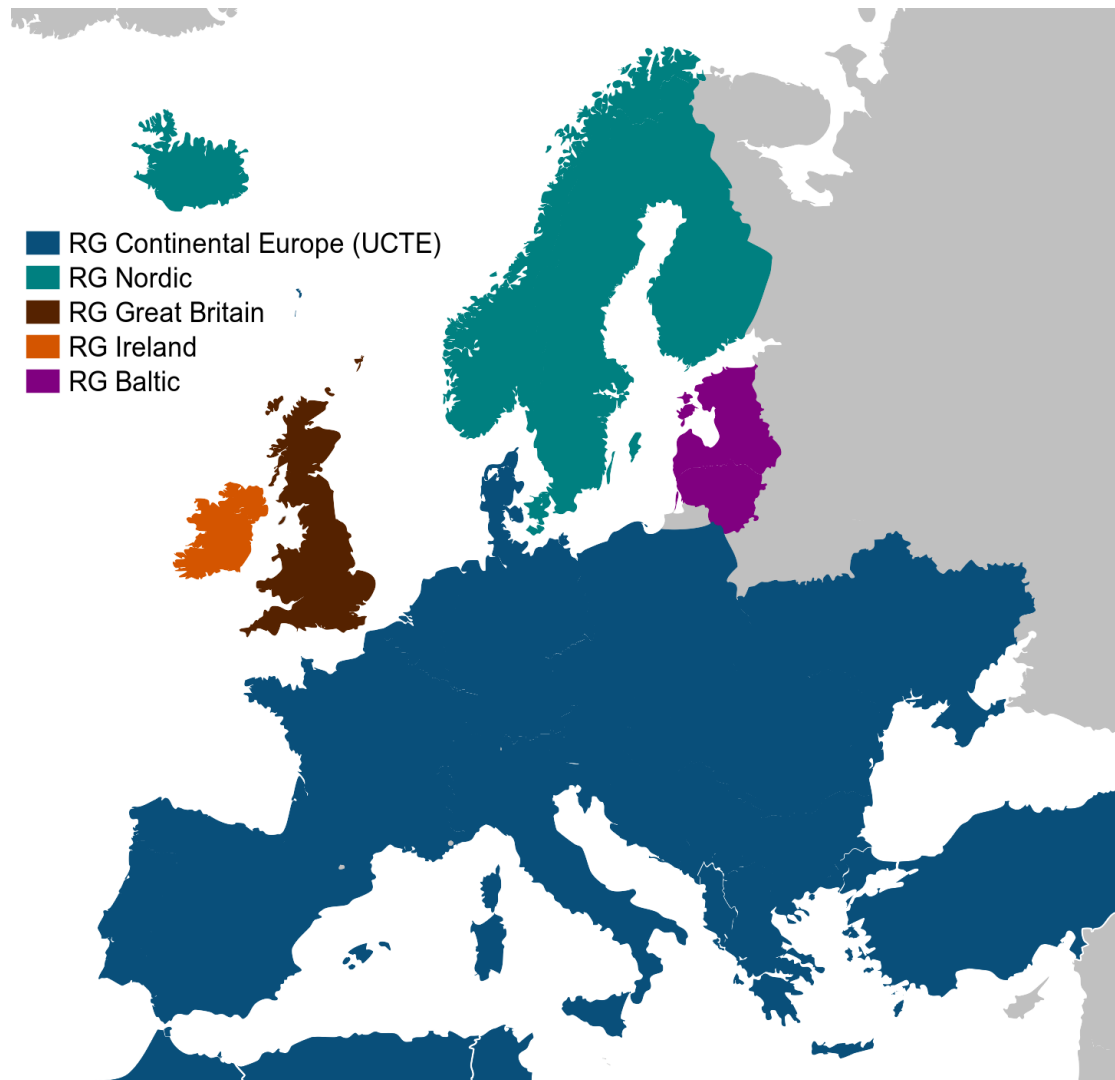


Figure 4.1.: Map of European Transmission System Operators Organisations (Regional Groups) Continental Europe, Nordic, Baltic, Great Britain and Ireland/Northern Ireland (former UCTE, UKTSOA, NORDEL, ATSOI, IPS/APS) [7].

With a brief introduction of the electrical grid, it would be interesting to have a look at a continental grid known as the **European Grid**.

4.4.2. The European Grid

The synchronous grid of Continental Europe⁴ is the largest synchronous electrical grid (by connected power) in the world. ⁴formerly known as the UCTE grid.

It is interconnected as a single phase-locked 50 Hz mains frequency electricity grid that supplies over 400 million customers in 24 countries, including most of the European Union.

In 2009, 667 GW of production capacity was connected to the grid, providing approximately 80 GW

of operating reserve margin.

4.4.3. The Super Grid

There are current proposal by the EU Commission to create a super-grid which would ultimately interconnect the various European countries and the regions around Europe's borders - including North Africa, Kazakhstan, and Turkey - with a high-voltage direct current (HVDC) power grid.

4.4.4. Feeding to the Grid

The grid itself provides the frequency and voltage reference as discussed previous with the EU grid shown as example with its set Voltage level and 50 Hz frequency. The inverter is programmed to feed current into its connection with the grid, in a direction that feeds power in, and a magnitude that looks like a negative resistance.

Imagine a resistor connected to the grid, i.e., a 3 kW kettle of roughly $20\ \Omega$. When the instantaneous grid voltage is 100 V, it draws 5 A, when it's 300 V, it draws 15 A. Using electronic control system, a load that **sensed** the instantaneous voltage can be created, and drew that amount of current. This would look like a resistor at its terminals, and they're called power factor corrected power supplies.

This can be done in reverse, sense the grid voltage, and feed current into the grid. That's how a grid-tie inverter sends power into the grid in a controlled fashion. To inject efficiently and safely into the grid, grid-tie inverters must accurately match the voltage, frequency and phase of the grid sine wave AC waveform.

4.4.5. Island Grid

A micro-grid is a local grid that is usually part of the regional wide-area synchronous grid but which can disconnect and operate autonomously [9]. It might do this in times when the main grid is affected by outages. This is known as islanding, and it might run indefinitely on its own resources. Island grids are an electrical power supply task with a small number of power generating plants and consumers. Island grids do not have a synchronous connection to a large network and therefore have to be able to provide all tasks necessary for long-lasting and safe operation on their own. Micro-grids are similar, but also have the capability to connect synchronously to a large network. Island grids are typically the result of geographical circumstances that render the connection to a large network costly or even impossible.

Micro-grids, in contrast, are designed to increase the security of supply in case the large network breaks down.

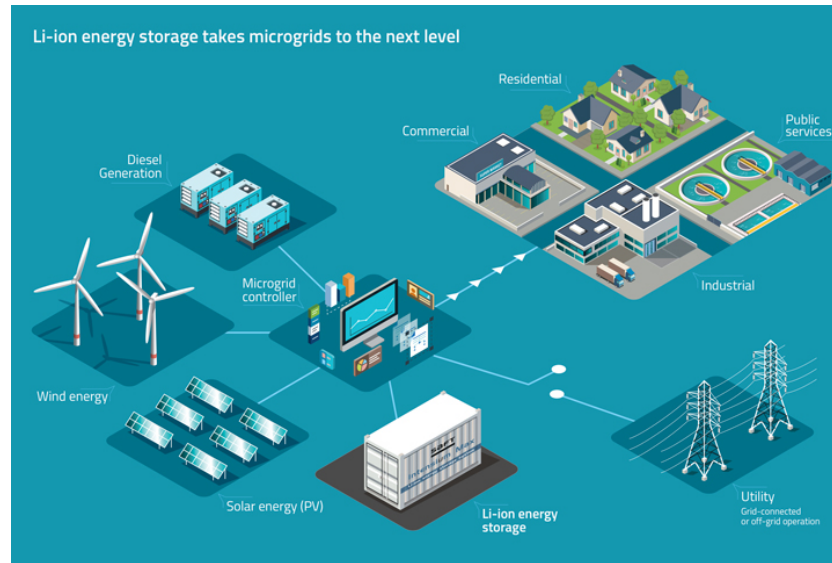


Figure 4.2.: One of the fundamentals of Microgrids is that they can become a localised source of energy for a commercial, industrial, or residential application, and they can disconnect from the main grid to provide a renewable source of energy.

4.4.6. Solar Panel Types

Solar panels are a popular choice for renewable energy generation. It is important to understand the different types of solar panels in order to make an informed decision for your energy needs. Here we will have a look at the key differences between monocrystalline, polycrystalline, and thin-film solar panels, highlighting their potential benefits and drawbacks.

Monocrystalline

Known for their high efficiency rates due to their single-crystal structure. The uniformity of the crystal structure allows for greater electron flow, resulting in higher power output. However, monocrystalline panels are also the most expensive option and can be less efficient in extreme temperatures. They typically have a lifespan of 25 to 30 years and provide a reliable source of clean energy.

Polycrystalline

Made from multiple silicon crystals, resulting in a lower efficiency compared to monocrystalline panels. However, they are more cost-effective to produce and perform better in high-temperature conditions. Polycrystalline panels have a slightly shorter lifespan of 20 to 25 years but still offer a reliable source of renewable energy.

Thin-film

Most light-weight and flexible option. They are made by depositing a thin layer of photovoltaic material onto a substrate, such as glass or metal. While thin-film panels have lower efficiency rates compared to monocrystalline and polycrystalline panels, they excel in low-light conditions and can be used in various applications. However, they have a shorter lifespan of around 10 to 20 years and may require more frequent replacement.

4.5. Lab Safety Rules

Before you begin, we have to come some lab safety rules.

Experiment lights can become very hot, **do not touch it**, except the switches at the back.

During grid testing, if something is not right, **press the emergency stop switch**.

4.6. Tasks

4.6.1. First Task

The 1st task is about the photovoltaic cells, consisting of two (2) panels in series, and their performance in parallel and serial configuration:

1. Look and understand the components you'll be working with and set up the experiment.

If you need help with the setup, ask your lab supervisor (Manuel Berger, Ph.D) for help.

2. Switch on the lighting equipment(s) and measure the radiant power on all the panels.

The dimension of a **single panel** is 1,125 m by 0,5 m.

3. Measure the no-load voltage and current of all panels.

a) You need to use multimeters for this part of the experiment.

4. Put a variable power resistor(s) as load, ranging from $0\ \Omega$ to $84\ \Omega$, to a single module and measure the consumed power to get the modules power with respect to the load. The connection should look roughly like below.

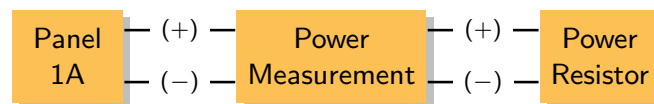


Figure 4.3.: Connection diagram for the task 1.3 with a single power resistor.

5. Redo the procedure for two modules in a parallel and four panels in serial configuration.

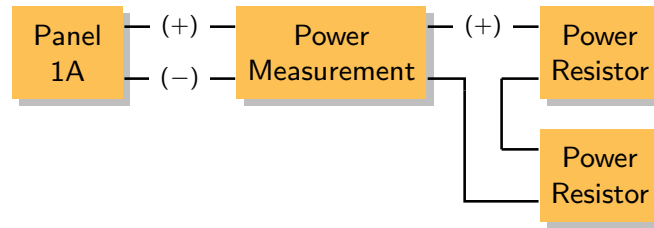


Figure 4.4.: Connection diagram for the task 1.3 with two power resistors.

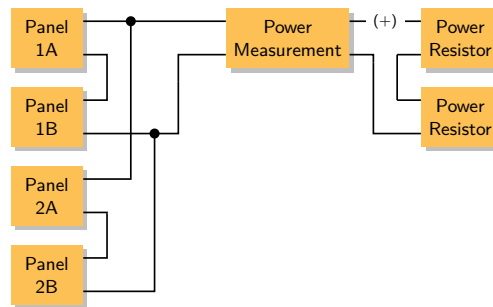


Figure 4.5.: Connection diagram for the task 1.3 with a single power resistor.

4.6.2. Second Task

The 2nd task is about **island grids (i.e., micro-grids)**:

1. Connect one module (i.e., two photovoltaic cells in series) to the grid-board for the island (i.e., micro-grid) configuration. Here the charger will have a load (R) ranging from $0\ \Omega$ to $42\ \Omega$.
2. Once the connection has been set up, run the experiment and measure the efficiency (η) of the charger using the connected power measurement devices.
3. Redo the task with an additional second module in parallel.

4.6.3. Third Task

The 3rd task is about **connecting the photovoltaic cells to the grid**.

1. Connect the modules to the board for the grid configuration (network activator, inverter, transformer).
2. **Plug the board to a power outlet**, run the experiment.
3. Estimate the efficiency (η) of the individual components and the whole setup.
 - a) This means the energy from the photons is input, and the energy fed to the grid is output.
4. Observe the flow of the energy.

What to Hand in

Your lab report, with your findings **must** be submitted to SAKAI no later than **2 weeks** after the experiment is concluded. Your lab report should roughly include the following points:

Topic	Description
Cover Page	The cover page needs to include: Title of the Experiment Student Name(s) Student Number(s) Module
Abstract	A short description of your work where you summarise your results in 100 - 200 words.
Introduction	The point in which you can hook the reader into reading your report. Start in with a broad picture and as you continue funnel the reader into what is relevant to your work. Here also include the literature research done for the lab preparation.
Materials and Methods	Explain to the reader what tools were used and give enough information for the work to be reproducible.
Results	Provide the reader with all the information gathered from the lab results and present them in a table of figure format.
Discussion and Conclusion	Considers whether the data you obtained support the hypothesis and discuss the ramifications of your results.
References	Present all the relevant references for your work. As the work is carried out in the electrical engineering, use the IEEE citation format.

Table 4.1.: A brief look on the stages of writing a lab report.

If you need a refresher or would like to learn more about scientific lab report writing, please look at the Appendix.

Part I.

Appendix

Part A

Lab Report Requirements

Table of Contents

A.1. Introduction	i
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A.1. Introduction

This section describes a general format for lab reports which you can adapt as needed. Lab reports are the most frequent kind of document written in the engineering discipline and can count for as much as 25% of a course yet little time or attention is devoted to how to write them well which this section hopes to improve, and to that end, this part has been written to be as topic agnostic as possible. It can also become a friction when each lecturer wants/require something a little different. Regardless of variations, however, the goal of lab reports remains the same:

... to document your findings and communicate their significance.

Keeping this in mind, we can describe the report's format and basic components. Knowing the pieces and purpose, we can then adapt to the particular needs of a course.

A good lab report does more than present data [10], [11], [12];

It demonstrates the writer's comprehension of the concepts behind the data.

Merely recording the expected and observed results is **NOT** sufficient to get a good grade as you should also identify **how** and **why** differences occurred, explain how they affected your experiment, and show your understanding of the principles the experiment was designed to examine.

A format, however helpful, cannot replace clear thinking and organised writing.

You still need to organise your ideas carefully and express them coherently. This report structure follows the University of Oxford Style guide [13] which can be found [here](#).

A.2. A Good Lab Report Structure

A good lab report usually requires a structure as such given below:¹

- | | |
|----------------------------|--------------------------------------|
| 1. Title Page, | 7. Discussion, |
| 2. Abstract, | 8. Conclusion, |
| 3. Introduction, | 9. References, |
| 4. Methods and Materials, | 10. Appendices, |
| 5. Experimental Procedure, | 11. Further Reading (if applicable). |
| 6. Results, | |

¹Of course this is a template as every course might require slight adjustments.

To have a good report for your experiments, you should be clear and concise while giving all the relevant information as possible. Therefore a good report, with all of its Appendices and images provided, including cover page, should **NOT** pass over thirty (30) pages. Let's look at these items in detail and elaborate on them, and of course show examples as well.

Title Page

The title page needs to contain the **name of the experiment** or **assignment**, the **names of lab partners or group members**, and the date. Titles should be straightforward, informative, and should **NOT** be unreasonably wrong (i.e., Not "Lab #4" but "Laboratory Report 3: Sample Analysis using the Debye-Sherrer Method").

The language will also play a role as the header and title format needs to be adjusted based on the language being used in teaching the said course.

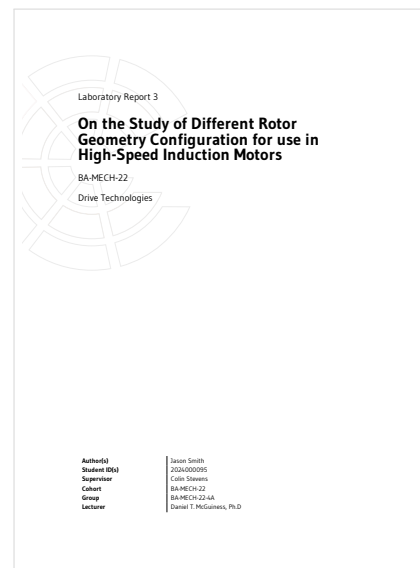


Figure A.1.: A good title page example for a report or an assignment. This title page was created using the `mcidoc` document class which can be found [here](#).

Therefore make sure the title page is formatted in German for German courses and English for the English courses.

Title page should also contain additional information, relevant to the lecture such as:

the degree, the program, cohort, group (if applicable), lecturer, lab supervisor.

Abstract

An abstract in any research paper or experimental work summarises four (4) essential aspects:

1. The purpose of the experiment²
2. Describing the key findings of the experiment
3. Their significance
4. Major conclusions

²sometimes expressed as the purpose of the report.

The abstract often also includes a brief reference to **theory** or **methodology**. The information should clearly enable readers to decide whether they need to read your whole report. The abstract should be one paragraph of 100–200 words.

The following example is 191 words [14].

Abstract

This experiment examined the effect of line orientation and arrowhead angle on a subject's ability to perceive line length, thereby testing the Müller-Lyer illusion. The Müller-Lyer illusion is the classic visual illustration of the effect of the surrounding on the perceived length of a line. The test was to determine the point of subjective equality by having subjects adjust line segments to equal the length of a standard line. Twenty-three subjects were tested in a repeated measures design with four different arrowhead angles and four line orientations. Each condition was tested in six randomised trials. The lines to be adjusted were tipped with

outward pointing arrows of varying degrees of pointedness, whereas the standard lines had inward pointing arrows of the same degree. Results showed that line lengths were overestimated in all cases. The size of error increased with decreasing arrowhead angles. For line orientation, overestimation was greatest when the lines were horizontal. This last is contrary to our expectations. Further, the two factors functioned independently in their effects on subjects' point of subjective equality. These results have important implications for human factors design applications such as graphical display interfaces.

As stated, the goal of an abstract is for reader to quickly assess your report and understand the overview of your work without reading the work itself. So think of this as a scientific elevator pitch. You need to sell your work to the reader in under 200 words.

Introduction

Once the abstract has sold your work to the reader, the next comes is the introduction. The introduction is **more narrowly focused** than the abstract. It states the objective of the experiment and provides the reader with background to the experiment³. State the topic of your report clearly and concisely, in one or two sentences. As an example please look below [14]:

The purpose of this experiment was to identify the specific element in a metal powder sample by determining its crystal structure and atomic radius. These were determined using the Debye-Sherrer (powder camera) method of X-ray diffraction.

³This is generally known as the *literature review* within the academic circle.

A good introduction is **NOT** just a sentence or two as its primary goal is to get the now interested reader up to speed with your work. To that end you should provides all the necessary background theory, previous research, or formulas the reader needs to know.

Please do not repeat the lab manual, but to show your own understanding of the problem and all the relevant information surrounding it.

For example, the introduction following the aforementioned example might describe the Debye-Sherrer method [15], and explain that from the diffraction angles the crystal structure can be found by applying Bragg's law [14], [16].

If the amount of introductory material seems to be a lot, consider adding subheadings such as: Theoretical Principles or Background.

Digression: A Note on Using the Verb Tense

Introductions often create difficulties as it is the place where careful attention must be made for using the correct tense. Below are two (2) points which should help you navigate the introduction:

- The experiment is already **finished**. Use the **past tense** when talking about the experiment.

"The objective of the experiment was ..."

- The report, the theory, and permanent equipment **still exist**. It is important for these statement to be written in **present tense**:

"The purpose of this report is ..."

"Bragg's Law for diffraction is ..."

"The scanning electron microscope produces micro-graphs ..."

Methods and Materials

This can usually be a simple list, or a table but make sure it is **accurate** and **complete**. An example is shown in **Table A.1**. Please note the table header is on top of the table.

Table A.1.: The part list used in the experiment A

Manufacturer	Model	Description
Anaheem Automation	BLY172S-24V-4000	Brushless DC Motor [24 V, 53 W]
Cypress	CY8CKIT-037	PSoC 4 Motor Control Evaluation Kit
Cypress	CY8CKIT-042	PSoC 4 Pioneer Kit
Keysight	DSOX1102A	Oscilloscope, 70 MHz, 2 GS0a s-

In some cases, you can simply direct the reader to a laboratory manual, standard procedure or technical documentation. As an example:

“Equipment was set up as in IEC 360004:08 manual ...”

Experimental Procedure

This section describes the process in **chronological order**. Using clear paragraph structure, explain all steps **in the order they actually happened**, not as they were supposed to happen.

If the lab supervisor says you can simply state that you followed the procedure in the manual, be sure you still document occasions when you did not follow that **exactly** [14].

“At step 4 we performed four repetitions instead of three, and ignored the data from the second repetition ...”

The goal here is to make your work **reproducible**. Another researcher should be able to duplicate your experiment.

Results

This section is usually dominated by calculations, tables, figures, codes, and algorithms. Regardless, you still need to state all significant results explicitly in verbal form. As an example [14]:

“Using the calculated lattice parameter gives, then, $R = 0,1244 \text{ nm}$...”

Graphics need to be clear, easily read, and must be labelled⁴. It is also worth mentioning that a good scientific report uses as much **vector** graphics as possible for their plots.

⁴e.g., **Figure 1.1:**
Input Frequency and
Capacitor Value

As an example, it is **NOT** a good practice to take a photo of the oscilloscope and put it into the report. You should interface with the Oscilloscope to retrieve the data and use your plotting software of choice⁵ to produce publishing quality plots. If the image cannot be produced in a vector format, it needs to have a minimum of 300 dpi as per publishing standard of IEEE.

⁵MATLAB, gnuplot,
pstricks, pgfplots, ...

An important strategy for making your results effective is to draw the reader's attention to them with a sentence or two, so the reader has a focus when reading the graph.

Regarding figures, avoid which are not mentioned in the report. If it is **NOT** mentioned in the report perhaps it is not vital.

In most cases, providing a sample calculation is sufficient in the report. Leave the remainder in an Appendix. Likewise, your raw data can be placed in an appendix as well with any major code you have used in your work no matter how trivial. Refer to Appendices as need arises, pointing out trends and identifying special features.

Discussion

This is the most important part of your report, as you show that you understand the experiment beyond the simple level of completing it.

Explain — Analyse — Interpret.

Some people like to think of this as the **subjective** part of the report. By that, they mean this is what is not readily observable. This part of the lab focuses on a question of understanding

"What is the significance or meaning of the results? "

To answer this question, use both aspects of discussion [14]:

Analysis	Interpretation
What do the results indicate clearly? What have you found? Explain what you know with certainty based on your results and draw conclusions:	What is the significance of the results? What ambiguities exist? What questions might we raise? Find logical explanations for problems in the data:
Since none of the samples reacted to the Silver foil test, therefore sulfide, if present at all, does not exceed a concentration of approximately 0.025 g/l. It is therefore unlikely that the water main pipe break was the result of sulfide-induced corrosion.	Although the water samples were received on 14 August 2000, testing could not be started until 10 September 2000. It is normally desirable to test as quickly as possible after sampling in order to avoid potential sample contamination. The effect of the delay is unknown.

Notice that, after the material is identified in the example above, the writer provides a justification. We know it is nickel because of its structure and size. This makes a sound and sufficient conclusion. Generally, this is enough; however, the conclusion might also be a place to discuss weaknesses of experimental design, what future work needs to be done to extend your conclusions, or what the implications of your conclusion are.

References

References include your lab manual and any outside reading you have done. As this is a mechatronics lab, your reference formatting should conform to IEEE citation standard. The citation standard can be seen [here](#) or you can look at the end of this lab report to see the standard as well.

Appendices

This section is usually reserved for content which does not particularly fit to any other section. Typically includes such elements as raw data, calculations, graphs pictures or tables which have not been included in the report itself. Each kind of item should be contained in a separate appendix. Make sure you refer to each appendix at least once in your report. For example, the results section might begin by noting [14]:

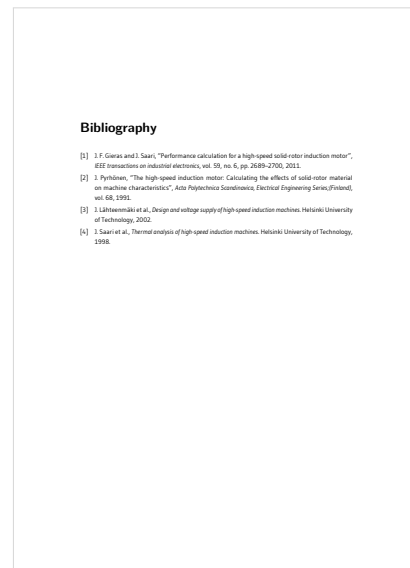
“Micro-graphs from the Scanning Electron Microscope are shown in Appendix A . . .”

A.3. Documentation Standard

The document should ideally be made in $\text{\LaTeX}$. For those who don't know, it is a type-setting language which allows you to create beautiful documents [17]. There are great resources to learn how to write a document in $\text{\LaTeX}$ such as Overleaf or the TeX stack-exchange and following books would give you more information than you could possibly need:⁶

- The $\text{\LaTeX}$ Companion (3rd Edition)
- The $\text{\LaTeX}$ Graphics Companion (2nd Edition)

The author of this section has also authored a $\text{\LaTeX}$ document class conforming to the MCI requirements for laboratory or assignment documentation which can be found [here](#). The class also includes a README to get you started.



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Figure A.2.: An example of a bibliography using BibTeX and IEEE citation standard. This bibliography entry was created using the `mcidoc` document class which can be found [here](#).

⁶Of course these are recommendations for people who want to understand $\text{\LaTeX}$ as a programming language rather than just a template generator.

Glossary

AC Alternating Current. 2, 3, 12–14, 23

DC Direct Current. 2, 3, 11, 12

IM Induction Machine. 1, 2, 5, 6, 10–12, 15, 16

PLC Programmable Logical Controller. 1–3, 7

PM Permanent Magnet. 12

SVC Static VAR Compensator. 14

SynM Synchronous Machine. 10–13

VFD Variable Frequency Drive. 2

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